

Distributed Nash Equilibrium Seeking and Privacy Preservation in Multi-Agent Systems with Unmodeled Dynamics

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Abstract

The Nash equilibrium seeking problem in the presence of unmodeled disturbances and privacy preservation is critical in which multiple agents participate together. This paper studies the Nash equilibrium seeking problem of multi-agent collective games and proposes a distributed Nash equilibrium seeking strategy for non-cooperative games based on gradient descent strategy and leaderless-consensus protocol. With Lyapunov stability analysis, the neighborhood where the behavior of the participants converges to the Nash equilibrium is analyzed. Unknown disturbances in the actual industrial environment are taken into account, and they are observed and suppressed through prescribed time extended state observer (PTESO). For privacy preservation, initial states of agents are decomposed to achieve the Nash equilibrium of the game. This provides a theoretical basis for the effective application of game theory methods in actual cyber-physical systems. Under given conditions, the proposed strategy can make the actions of participants converge to a Nash equilibrium. Numerical examples verify the effectiveness of the optimization strategy.

CCS Concepts

- Security and privacy → Information-theoretic techniques;
- Computing methodologies → Distributed algorithms.

Keywords

Nash equilibrium, Distributed strategy, Privacy preservation, PTESO, Multi-agent systems

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1 Introduction

As artificial intelligence and communication technologies advance, multi-agent systems (MASs) have become a research focus across various fields (see [2, 3, 9, 17, 20, 21, 25] for more comprehensive reviews). These systems consist of agents with unique sensing, computing, and acting abilities, working together to achieve collective control goals [6, 10, 22]. Researchers have integrated game theory to model agent behavior and analyze objective functions, loss

functions, and strategic actions, aiming to find the game model's optimal solution, the Nash equilibrium, thus enhancing system performance [7, 8, 19]. However, models in real systems are often more complex, influenced by unmodeled dynamics and malicious entities. The effective application of game theory to real-world cyber-physical systems remains a challenging area for exploration and research.

In actual systems, since the communication channels through which agents exchange information are not protected, they are vulnerable to malicious attacks by hostile agents. Some malicious attacks will steal information while blocking and interfering with the process of information transmission, affecting the process of agents' strategy. Therefore, hiding the state of an agent from being stolen to achieve privacy protection has become a widely studied issue. Unmodeled disturbances that exist in actual physical systems are an unavoidable problem, including changes in measurement noise, environmental effects, and other factors, and there are often internal state variables that are difficult to measure in multi-agent systems. In view of this, it is usually necessary to perform state estimation on these unknown perturbation structure observers in the system and implement corresponding compensation to offset the impact of the perturbation.

Authors in [2, 14] propose an asymmetric projection algorithm for finite player games and proved that it converges to the Nash equilibrium point. Research in [12, 20] employs gaming techniques to simulate and scrutinize the system's energy usage. By facilitating communication among neighboring users, it's possible to gauge the aggregate energy expenditure. The document suggests a methodical approach for Nash equilibrium seeking and examines the strategy's ability to converge. The local convergence results are given, and finally the convergence of the algorithm is analyzed using singular perturbation analysis and Lyapunov stability. The literature [25] considers the impact of the undirected communication topology structure on consistent network systems in a noisy environment, and uses the expected steady-state variance of the state to consider the system robustness in a noisy environment. Literature [23] and [11] studies a class of nonlinear dynamic system aggregation games subject to uncertainty disturbances, which allows participants in the game not only to seek for the Nash equilibrium of the aggregation game, but also to suppress the influence of external disturbances. However, the performance of the observers constructed in the above studies is limited. Moreover, the state of the error system cannot eventually converge to 0, and the convergence time cannot be guaranteed. In [13, 21], an extended state observer is constructed that can converge within a specified time. This observer can obtain better observation performance, and all error states of the system converge to 0 within a given time. On this basis, we present a downgrade compensation scheme for the perturbation. The rest of this paper is organized as follows.

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Section 2 provides problem formulation and some preliminaries including graph theory, model, and state decomposition mechanism. Section 3 presents our main results, which provides a corresponding proof that the substates after state decomposition can still converge to the neighborhood of the Nash equilibrium in the presence of unmodeled dynamics of the system, and prescribed time extended state observer (PTESO) convergence is proven by the Lyapunov theory. A numerical simulation is shown in Section 4. Finally, Section 5 concludes this paper.

The main contributions of this paper are summarized below.

- (1) A distributed Nash equilibrium seeking framework is developed for games with unmodeled external disturbances. By integrating a gradient descent strategy with a second-order PTESO, the proposed method compensates for unknown disturbances and guarantees convergence to the Nash equilibrium under directed communication graphs.
- (2) A privacy-preserving mechanism based on state decomposition is integrated into the algorithm. Only one substate is transmitted over the network, while the complementary substate and internal weights remain private. An indistinguishability-based privacy definition is formulated, and it is shown that the initial state of each agent cannot be uniquely inferred by hostile agents.

2 Problem Formulations

2.1 Graph Theory Notions

A graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ can be represented by a set of vertices $\mathcal{V} = \{1, 2, \dots, N\}$ and a set of edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. Specifically, a multi-agent system can be abstracted as a graph, where the vertices of the graph represent an agent and the edges between the vertices represent the communication between the agents. If the edges set $\mathcal{E}$ are constantly changing, then we say that the graph is time-varying. For an agent i , we define the agents that send data to it as in-neighbors, and the agents that receive information from it as out-neighbors. Thus, we have two sets $\mathcal{V}_i^- = \{j \mid j \text{ is the agent that sends data to } i\}$ and $\mathcal{V}_i^+ = \{j \mid j \text{ is the agent that receive data from } i\}$.

2.2 Nash Equilibrium

In this paper, we consider an MAS with N agents. Among them, the agent i intends to

$$\begin{aligned} & \min_{\mathbf{x}_i} f_i(\mathbf{x}) \\ & \text{subject to } \dot{\mathbf{x}}_i = \mathbf{u}_i + \mathbf{d}_i(t), \end{aligned} \quad (1)$$

where $i \in \mathcal{V}$, $\mathbf{x}_i$, $\mathbf{u}_i$ and $f_i(\mathbf{x})$ are the action, the control input, and the objective function of player i , respectively and $\mathbf{d}_i(t)$ represents unmodeled dynamics term existing in the system. Define $\mathbf{x} = [x_1, x_2, \dots, x_N]^T$ is the set of actions of the players. For convenience of notation, define $\mathbf{x}_i = [x_1, x_2, \dots, x_{i-1}, x_{i+1}, \dots, x_N]^T$. Following the above notations, Nash equilibrium is defined below.

DEFINITION 1 ([21]). *If each player $i \in \mathcal{V}$ for all $\mathbf{x}_i \in \mathbf{x}$ all hold the following inequality*

$$f_i(\mathbf{x}_i^*, \mathbf{x}_i^*) \leq f_i(\mathbf{x}_i, \mathbf{x}_i^*). \quad (2)$$

Then $\mathbf{x}^ = (\mathbf{x}_1^*, \mathbf{x}_2^*)$ is called the Nash equilibrium of the game $\Gamma = \{\mathcal{V}, \mathbf{x}, f\}$.*

As $f_i(\mathbf{x})$ is a continuous function and quadratically differentiable, the following assumptions are commonly used to solve the Nash equilibrium.

ASSUMPTION 1. *If there is a Nash equilibrium $\mathbf{x}^* = [x_1^*, x_2^*, \dots, x_N^*]^T$, $i \in \mathcal{V}$, for a continuously differentiable function, then it needs to satisfy*

$$\begin{cases} \frac{\partial f_i}{\partial \mathbf{x}_i}(\mathbf{x}^*) = 0 \\ \frac{\partial^2 f_i}{\partial \mathbf{x}_i^2}(\mathbf{x}^*) > 0, \end{cases} \quad (3)$$

which means its convexity and strong monotonicity.

ASSUMPTION 2. *For a matrix*

$$B = \begin{bmatrix} \frac{\partial^2 f_1}{\partial \mathbf{x}_1^2}(\mathbf{x}^*) & \frac{\partial^2 f_1}{\partial x_1 \partial x_2}(\mathbf{x}^*) & \dots & \frac{\partial^2 f_1}{\partial x_1 \partial x_N}(\mathbf{x}^*) \\ \frac{\partial^2 f_2}{\partial x_2 \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f_2}{\partial \mathbf{x}_2^2}(\mathbf{x}^*) & \dots & \frac{\partial^2 f_2}{\partial x_2 \partial x_N}(\mathbf{x}^*) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f_N}{\partial x_N \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f_N}{\partial x_N \partial x_2}(\mathbf{x}^*) & \dots & \frac{\partial^2 f_N}{\partial \mathbf{x}_N^2}(\mathbf{x}^*) \end{bmatrix},$$

inequality $|\frac{\partial^2 f_i}{\partial \mathbf{x}_i^2}(\mathbf{x}^)| > \sum_{j=1, j \neq i}^N |\frac{\partial^2 f_i}{\partial x_i \partial x_j}(\mathbf{x}^*)|$ is always true. That is, the matrix is strictly dominant and is a non-singular matrix.*

Assuming that (3) is satisfied, the matrix B can be a Hurwitz matrix, and the corresponding Nash mean the balance is independent.

2.3 Models

Assume that each agent i performs synchronous interactions by passing their state values to their neighbors in the network. Each regular agent updates its value over time according to a prescribed rule. The update rule when the unmodeled dynamics is considered for agent i is designed as

$$\dot{\mathbf{x}}_i = \mathbf{u}_i(\mathbf{x}_i(t), \mathbf{x}_j(t)) + \mathbf{d}_i(t), \quad (4)$$

where $\mathbf{u}_i$ is a pre-designed control algorithm, and $\mathbf{d}_i(t)$ is unmodeled dynamics.

As for adversary model, suppose that there exists hostile nodes out of graph $\mathcal{G}$ who intend to steal privacy, or the initial state of certain agents. Define the properties of these external hostile nodes

- (1) Collaborate with each other.
- (2) Have the ability to obtain all information transmitted on the communication channels in graph $\mathcal{G}$, and to know the algorithm structure and communication topology.
- (3) Do not have the ability to access the internal sub-states or the private decomposition parameters, such as the initial segmentation and internal coupling weights.
- (4) Regular agents cannot require any information from them.

2.4 Privacy-Preserving State Decomposition Mechanism

Given our adversarial model, transmitting the agent's state in plaintext could expose the agent's sensitive information to adversarial agents.

This paper builds on the state decomposition method of [16] and considers unmodeled dynamics in MAS. In this method, the

agent's state x_i is decomposed into two sub-states x_i^a, x_i^b by a private decomposition function at the initial time instant.

Note that the substate x_i^a which replaces actual state communicates with other agents while the private substate x_i^b only interacts with x_i^a . Assuming that there is bidirectional communication between x_i^a and x_i^b , then their weights are $\omega_{i,21}, \omega_{i,12}$, satisfying $\eta \leq \omega_{i,21}, \omega_{i,12} < 1$, $0 < \eta < 1$.

First, the regular agent i receives the substate x_j^a from its in-neighbors $\mathcal{V}_i^-$ and transmit its substate x_i^a to the out-neighbors $\mathcal{V}_i^+$. The update laws for two substates are

$$\begin{cases} \dot{x}_i^a(t) = u_i(t) - \omega_{i,21}(t)g_i(t)z_i(t) \\ \dot{x}_i^b(t) = \omega_{i,12}(t)z_i(t), \end{cases} \quad (5)$$

where substates error $z_i(t) = x_i^a(t) - x_i^b(t)$, and the term $g_i(t)$ is a time-varying gain associated with the internal substates error

$z_i(t)$. Define $g_i(t) = \begin{cases} 1, & |z_i(t)| \leq \gamma_t, \\ \gamma_t/|z_i(t)|, & \text{otherwise.} \end{cases}$, where γ_t is a non-negative, monotonically decreasing function satisfying $\lim_{t \rightarrow \infty} \gamma_t = 0$.

Two substates and actual state of regular agents ultimately converge to the same value. Based on this state decomposition mechanism with unmodeled dynamics, the initial state can be effectively protected.

However, the unmodeled dynamics problem which may have a negative impact on the multi-agent system has not been considered. Based on this, the PTESO algorithm is introduced in Section 3.1.

3 Main Results

3.1 Second-Order PTESO Design

The original PTESO in [5] is constructed as a third-order observer to estimate both the disturbance and its derivative for second-order system dynamics. However, estimating the disturbance derivative is unnecessary since the agent dynamics considered in our work 4 is first-order. Therefore, a simplified second-order PTESO is designed by removing the derivative estimation layer, which is constructed as

$$\begin{cases} \varphi_{i1} = x_i^a - \hat{x}_i^a \\ \dot{\hat{x}}_i = u_i + \hat{d}_i + \mathcal{T}\zeta\Xi l_1\varphi_{i1} + (1 - \mathcal{T})k_1\Gamma^{\frac{1}{2}}[\varphi_{i1}]^{\frac{1}{2}} \\ \dot{\hat{d}}_i = \mathcal{T}\zeta^2\Xi^2 l_2\varphi_{i1} + (1 - \mathcal{T})k_2\Gamma[\varphi_{i1}]^0, \end{cases} \quad (6)$$

where $\hat{x}_i^a$ is the estimate of x_i^a , $\hat{d}_i$ is the estimate of unmodeled dynamics d_i ; $\Xi = \sec^2(\frac{\pi}{2}\frac{t}{\bar{t}_f})$, $t \in [0, \bar{t}_f]$, and $\bar{t}_f$ is the prescribed convergence time for the PTESO; Γ is a Lipschitz constant to be designed, which satisfies $\Gamma \geq \bar{d}_{i\max}$; $k_{1,2}$ are the parameters to be designed, which satisfy $k_2 > \Gamma$, $k_1 = \alpha k_2^{\frac{1}{2}}$ [24], and α is a constant; $[\cdot]^p = |\cdot|^p \text{sign}(\cdot)$; $\mathcal{T}$ is the time switching function of the PTESO satisfying $\mathcal{T} = \begin{cases} 1, & t \in [0, \bar{t}_f], \\ 0, & t \in [\bar{t}_f, +\infty) \end{cases}$.

Then the error system yields

$$\begin{cases} \dot{\varphi}_{i1} = \varphi_{i2} - \mathcal{T}\zeta\Xi l_1\varphi_{i1} - (1 - \mathcal{T})k_1\Gamma^{\frac{1}{2}}[\varphi_{i1}]^{\frac{1}{2}} \\ \dot{\varphi}_{i2} = \dot{d}_i - \mathcal{T}\zeta^2\Xi^2 l_2\varphi_{i1} - (1 - \mathcal{T})k_2\Gamma[\varphi_{i1}]^0, \end{cases} \quad (7)$$

where $\varphi_{i2} = d_i - \hat{d}_i$.

REMARK 1. The prescribed-time convergence of the PTESO is guaranteed when $\Gamma \geq \bar{d}_{i\max}$. However, if Γ is underestimated, the strict prescribed-time property may be weakened while the estimation error remains ultimately bounded. Choosing a larger Γ preserves the theoretical guarantee but may increase transient peaking and noise sensitivity. Therefore, the prescribed time t_f in practice should be selected according to practical factors like sampling and actuator constraints to balance convergence speed and robustness.

The following lemmas are necessary for our result.

LEMMA 1 ([5]). Suppose a time-varying function $h(t)$, which satisfies the property $h(\bar{t}_f) = +\infty$. There exists a real-valued continuously differentiable function $V(\cdot) : \mathbb{R} \times \mathbb{R}^n$ satisfying $V(x_i^a(t)) = 0 \Leftrightarrow x_i^a(t) = 0$, $\bar{k} > 0$ and $\bar{m} > 0$ such that $\dot{V}(x(t)) \leq -\bar{k}hV(x(t)) + \bar{m}h$ holds, then $\lim_{t \rightarrow \bar{t}_f} V(t)$ is bounded.

LEMMA 2 ([1]). Suppose the matrixes $S \in \mathbb{R}^{n \times n}$, $H \in \mathbb{R}^{n \times n}$ and

$$\bar{C} \in \mathbb{R}^{1 \times n} \text{ hold the following form } S = \begin{bmatrix} 0 & \sigma_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{n-1} \\ 0 & 0 & \cdots & 0 \end{bmatrix} \text{ with}$$

σ_k being 0 or 1, $k = 1, \dots, n-1$, $H = \text{diag}\{n, n-1, \dots, 1\}$ and $\bar{C} = [1, 0, \dots]$. For any constant γ , there exists a positive definite matrix $P \in \mathbb{R}^{n \times n}$ and a vector $R \in \mathbb{R}^{1 \times n}$ such that $(S - \bar{R}\bar{C})^T P + P(S - \bar{R}\bar{C}) \leq S + S^T - \gamma I_n$ and $\mu_1 I_n \leq PH + HP \leq \mu_2 I_n$, where μ_1 and μ_2 are positive constants, and I_n is an n -dimensional identity matrix.

LEMMA 3 ([4]). Given a continuously differentiable function $V_Y = \alpha_1 \gamma_1(v_1, v_2) + \alpha_2 \frac{1}{3}|v_2|^3$, where $\alpha_1 > 0$, $\alpha_2 > 0$, $v_1 = \frac{\varphi_{i1}}{\Gamma}$, $v_2 = \frac{\varphi_{i2}}{k_1 \Gamma}$, $\gamma_1(v_1, v_2) = \frac{2}{3}|v_1|^{\frac{3}{2}} - v_1|v_2|^{\frac{1}{2}} + \frac{1}{3}|v_2|^{\frac{3}{2}}$. For V_Y , there exists a parameter $\kappa > 0$ satisfying $\dot{V}_Y \leq -\kappa V_Y^{\frac{2}{3}}$.

Next, we will prove that the simplified second-order PTESO (6) can still apply to Theorem 1 in [24].

THEOREM 1. In system (5), when equipped with the designated PTESO (6), the error system's states (7) are guaranteed to converge before the prescribed time $\bar{t}_f$. Post convergence, they will remain at zero for $t > \bar{t}_f$.

PROOF. 1) Considering $t < \bar{t}_f$, the error system (7) yields

$$\begin{cases} \dot{\varphi}_{i1} = \varphi_{i2} - \mathcal{T}\zeta\Xi l_1\varphi_{i1} \\ \dot{\varphi}_{i2} = \dot{d}_i - \mathcal{T}\zeta^2\Xi^2 l_2\varphi_{i1}, \end{cases} \quad (8)$$

which can be rewritten as

$$\dot{\varphi}_i = A\varphi_i - \Psi LC\varphi_i + W, \quad (9)$$

where $\varphi_i = [\varphi_{i1}, \varphi_{i2}]^T$, $\Psi = \text{diag}\{\zeta\Xi, \zeta^2\Xi^2\}$, $L = [l_1, l_2]^T$, $C = [1, 0]$, $W = [0, \dot{d}_i]^T$, and the matrix A satisfies $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. For the ease of convergence analysis, introducing the state transformation $\tilde{\varphi} = \tilde{\Psi}\varphi_i$, and then get $\dot{\tilde{\varphi}} = \tilde{\Psi}\dot{\varphi}_i + \dot{\tilde{\Psi}}\varphi_i$ where $\tilde{\Psi} = \text{diag}\{\zeta^2\Xi^2, \zeta\Xi\}$. Accordingly, $\tilde{\Psi} = \rho\Xi D\tilde{\Psi}$ and $\varphi_i = \tilde{\Psi}^{-1}\tilde{\varphi}$, where $D = \text{diag}\{2, 1\}$, $\rho = \frac{\pi}{2\bar{t}_f} \sin(\frac{\pi}{2}\frac{t}{\bar{t}_f}) \leq \frac{\pi}{2\bar{t}_f}$. Then the equation (9) is rearranged into $\dot{\tilde{\varphi}} = \rho\Xi D\tilde{\Psi}(\tilde{\Psi}^{-1}\tilde{\varphi}) + \tilde{\Psi}[A(\tilde{\Psi}^{-1}\tilde{\varphi})] - \Psi LC(\tilde{\Psi}^{-1}\tilde{\varphi}) + W$.

It is easy to obtain that $\tilde{\Psi}A\tilde{\Psi}^{-1} = \Xi\Xi\zeta A$ and $\tilde{\Psi}\Psi LC\tilde{\Psi}^{-1} = \Xi\Xi\zeta LC$. Finally we obtain that $\tilde{\Phi} = \rho\Xi D\tilde{\Phi} + \Xi\Xi\zeta(A - LC)\tilde{\Phi} + \tilde{\Psi}W$.

Consider a Lyapunov function candidate as $V_{\tilde{\Phi}} = \tilde{\Phi}^T P \tilde{\Phi}$, where P is positive definite symmetric matrix. With the help of Rayleigh's inequalities, it is rearranged as $\lambda_{\min}(P)\|\tilde{\Phi}\|^2 \leq V_{\tilde{\Phi}} \leq \lambda_{\max}(P)\|\tilde{\Phi}\|^2$. With Lemma 2, the time derivative of $V_{\tilde{\Phi}}$ yields

$$\begin{aligned} \dot{V}_{\tilde{\Phi}} &\leq \bar{\nu}\rho\Xi\|\tilde{\Phi}\|^2 + 2\Xi\Xi\zeta\|A\|\|\tilde{\Phi}\|^2 - \Xi\Xi\zeta\gamma\|\tilde{\Phi}\|^2 \\ &\quad + 4\bar{p}\Xi\|\dot{d}_i\|\|\tilde{\Phi}\|, \end{aligned} \quad (10)$$

where $\bar{p} = \max\{|p_2|, |p_3|, |p_4|\}$ is a bounded number, and p_i for $i = 2, 3, 4$ is the elements of matrix $P = \begin{bmatrix} p_1 & p_2 \\ p_3 & p_4 \end{bmatrix}$. Apply Young's inequality, choose $\epsilon > 0$, $\zeta \geq \frac{\bar{\nu}\pi}{2\bar{t}_f} + \epsilon$, $\gamma \geq 1 + 2\|A\|$ and define $\varrho = \zeta(\gamma - 2\|A\| - 1) > 0$ the following inequality holds $\dot{V}_{\tilde{\Phi}} \leq -\vartheta\Xi V_{\tilde{\Phi}} + \frac{8\Xi}{\epsilon}(\bar{p}\|\dot{d}_i\|)^2$, where $\vartheta = \frac{\epsilon}{\lambda_{\max}(P)} > 0$.

With the help of Lemma 1, one has $\lim_{t \rightarrow \bar{t}_f} V_{\tilde{\Phi}}(t)$ is bounded. With $\lim_{t \rightarrow \bar{t}_f} \|\tilde{\Psi}(t)^{-1}\| = 0$, it indicates that $\lim_{t \rightarrow \bar{t}_f} \|\Phi_i(t)\| = 0$, and the observation error converges to $\mathbf{0}$ during the prescribed interval $[0, \bar{t}_f]$.

2) For $t \in [\bar{t}_f, +\infty)$, the error system can be rewritten as

$$\begin{cases} \dot{\phi}_{i1} = \varphi_{i2} - k_1\Gamma^{\frac{1}{2}}[\varphi_{i1}]^{\frac{1}{2}} \\ \dot{\phi}_{i2} = \dot{d}_i - k_2\Gamma[\varphi_{i1}]^0. \end{cases} \quad (11)$$

From the result in [4], we choose a positive definite function

$$V_Y = \alpha_1\gamma_1(v_1, v_2) + \alpha_2\frac{1}{3}|v_2|^3, \quad (12)$$

where arbitrary $\alpha_{1,2} > 0$, $v_1 = \frac{\varphi_{i1}}{\Gamma}$, $v_2 = \frac{\varphi_{i2}}{k_1\Gamma}$ and $\gamma_1(v_1, v_2)$ satisfies $\gamma_1(v_1, v_2) = \frac{2}{3}|v_1|^{\frac{3}{2}} - v_1[v_2]^{\frac{1}{2}} + \frac{1}{3}|v_2|^{\frac{3}{2}}$.

With Lemma 3 and (12), if the parameters are chosen as $k_2 > \Gamma \geq \dot{d}_{\max}$, $k_1 = \alpha k_2^{\frac{1}{2}}$, there exists a positive constant κ such that the time derivative of (12) yields

$$\dot{V}_Y \leq -\kappa V_Y^{\frac{2}{3}} \leq 0. \quad (13)$$

Thus, from (12), (13) and the result $\lim_{t \rightarrow \bar{t}_f} \|\Phi_i(t)\| = 0$, it is easy to obtain that the LF candidate satisfies $\lim_{t \rightarrow \bar{t}_f} V_Y = 0$. Thus, the LF candidate V_Y is going to remain on $V_Y = 0$ after the time instant $\bar{t}_f$, i.e., $\|\Phi_i\| \equiv 0$ is still maintained in the subsequent period. The proof is completed. $\square$

3.2 Distributed Nash Equilibrium Seeking System Formulation

This chapter mainly studies distributed game problems. The goal of this type of problem is to find the Nash equilibrium solution of the game so that all players can reach their optimal values. Since players can only exchange information with neighbors connected on the topological graph, it is impossible to directly obtain the information of all players. Therefore, we assume that player i will estimate the behavior of player j , expressed as $\hat{y}_{ij}$. The set of estimates made by player i regarding the actions of other players is expressed as $y_i = [\hat{y}_{i1}, \hat{y}_{i2}, \dots, \hat{y}_{iN}]^T \in R^N$.

Combining the gradient descent method and the consistency protocol, the update strategy adopted by the system can be expressed

as

$$\begin{cases} u_i = -\frac{\partial f_i}{\partial x_i}(y_i) - \hat{d}_i, & i \in \mathcal{V} \\ \hat{y}_{ij} = -\left(\sum_{k=1}^N w_{ik}(y_{ij} - y_{kj}) + w_{ij}(y_{ij} - x_j^a)\right), & j \in \mathcal{N}_i^-. \end{cases} \quad (14)$$

Define

$$\xi_i = x_i^a - x_i^*, \eta_{ij} = y_{ij} - x_j^a, \quad (15)$$

where x_i^{a*} is defined in Definition 1, then

$$\begin{cases} \dot{\xi}_i = \dot{x}_i^a = -\frac{\partial f_i}{\partial x_i}(y_i) + \varphi_{i2} - \omega_{i,21}(t)g_i(t)z_i(t) \\ \dot{\eta}_{ij} = \dot{y}_{ij} - \dot{x}_j^a \\ = -\theta_{ij}\left(\sum_{k=1}^N a_{ik}(\eta_{ij} - \eta_{kj}) + a_{ij}\eta_{ij} - \dot{\xi}_j\right). \end{cases} \quad (16)$$

Thus

$$\begin{cases} \dot{\xi} = -\left[\frac{\partial f_i}{\partial x_i}\right]_{\text{vec}} + \varphi_2 + \gamma \\ \dot{\eta} = -\theta\bar{\Theta}(\mathcal{L} \otimes I_{N \times N} + \mathcal{A}_0)\eta - \mathbf{1}_N \otimes \dot{\xi}, \end{cases} \quad (17)$$

where $\eta = \text{diag}\{\eta_{ij}\}$, $\xi = [\xi_i]_{\text{vec}}$, $\varphi_2 = [\varphi_{i2}]_{\text{vec}}$, $\bar{\Theta} = \text{diag}\{\bar{\theta}_{ij}\}$, $\gamma = [-\omega_{i,21}(t)g_i(t)z_i(t)]_{\text{vec}}$, $\mathcal{A}_0 = \text{diag}\{a_{ij}\}$, and $\mathbf{1}_{N \times N}$ is an identity matrix of dimension $N \times N$.

3.3 Stability Analysis

THEOREM 2. Consider the closed-loop system formed by (5) and (14) with the PTESO (6), which yields the aggregate error dynamics (17). Suppose that Assumptions 1-2 hold. Then, there exist constants $\theta^* > 0$ such that for any $\theta \in (\theta^*, \infty)$, $\|\Phi(t)\| \rightarrow 0$ as $t \rightarrow \infty$. Consequently, the substate x_i^a converges to the Nash equilibrium x_i^* , i.e.,

$$\lim_{t \rightarrow \infty} x_i^a(t) = x_i^*, \quad \forall i \in \mathcal{V}. \quad (18)$$

PROOF. First, we choose

$$V = \eta^T P \eta + \frac{1}{2} \xi^T \xi, \quad (19)$$

and P is a symmetric positive-definite matrix that satisfies

$$P(\bar{\Theta}(\mathcal{L} \otimes I_{N \times N} + \mathcal{A}_0)\eta) + (\bar{\Theta}(\mathcal{L} \otimes I_{N \times N} + \mathcal{A}_0)\eta)^T P = Q, \quad (20)$$

where Q is a symmetric positive-definite matrix.

Then, by utilizing Assumption 1 and Assumption 2, there exist positive constants $l_i, \forall i \in \{1, 2, 3, 4\}$ such that

$$\begin{aligned} \dot{V} &\leq -\theta\lambda_{\min}(Q)\|\eta\|^2 - 2\eta^T P(\mathbf{1}_N \otimes \dot{\xi}) \\ &\quad - m\|\xi\|^2 - \xi^T \left[\frac{\partial f_i}{\partial x_i}(y_i) - \frac{\partial f_i}{\partial x_i}(x) \right] + \xi^T(\varphi_2 + \gamma_t \omega_{21}) \\ &\leq -\theta\lambda_{\min}(Q)\|\eta\|^2 + l_1\|\eta\|^2 + l_2\|\eta\|\|\xi\| \\ &\quad + l_3\|\eta\|\|\varphi_2 + \gamma_t \omega_{21}\| - m\|\xi\|^2 \\ &\quad + l_4\|\xi\|\|\eta\| + \|\xi\|\|\varphi_2 + \gamma_t \omega_{21}\|, \end{aligned} \quad (21)$$

where $\omega_{21} = [\omega_{i,21}]_{\text{vec}}$.

Define $A_1 = \begin{bmatrix} \theta\lambda_{\min}(Q - l_1) & -\frac{l_2+l_4}{2} \\ -\frac{l_2+l_4}{2} & m \end{bmatrix}$. Then

$$\dot{V} \leq -\lambda_{\min}(A_1)\|\bar{\Phi}\|^2 + (l_3 + 1)\|\bar{\Phi}\|\|\varphi_2 + \gamma_t \omega_{21}\|, \quad (22)$$

where $\lambda_{\min}(A_1) > 0$ and is given as $\theta > \theta^* = \frac{(l_2+l_4)^2+4ml_1}{4m\lambda_{\min}(Q)}$.

Hence, by choosing $\theta > \theta^*$, $\dot{V} \leq -\frac{\lambda_{\min}(A_1)}{2} \|\bar{\phi}\|^2$ holds for any $\|\bar{\phi}\| \geq \frac{2(l_3+1) \sup_{t \geq t_0} \|\varphi_2 + \gamma_t \omega_{21}\|}{\lambda_{\min}(A_1)}$. Thus, $\|\bar{\phi}\| \rightarrow \frac{2(l_3+1) \sup_{t \geq t_0} \|\varphi_2 + \gamma_t \omega_{21}\|}{\lambda_{\min}(A_1)}$, $t \rightarrow \infty$. Recall that φ_2 is the estimation error of the observer for the unknown dynamics, which in the previous section has been proven to converge to 0 when $t \geq \bar{t}_f$, and γ_t is a non-negative monotonically decreasing function satisfying $\gamma_t \rightarrow 0$, $t \rightarrow \infty$.

Finally, it is obtained that

$$\|\bar{\phi}\| \rightarrow 0 \text{ as } t \rightarrow \infty. \quad (23)$$

The proof is completed. $\square$

Now it is obtained that $t \rightarrow \infty, x_i^a \rightarrow x_i^*, \forall i \in \mathcal{V}$. Next it will be proved that $t \rightarrow \infty, x_i^b \rightarrow x_i^*, \forall i \in \mathcal{V}$.

THEOREM 3. *Under the condition of Theorem 2, the two sub-states and the actual state of regular agents can achieve Nash equilibrium, despite the presence of unknown disturbances, i.e.,*

$$\lim_{t \rightarrow \infty} x_i(t) = \lim_{t \rightarrow \infty} x_i^a(t) = \lim_{t \rightarrow \infty} x_i^b(t) = x_i^* \quad \forall i \in \mathcal{V}. \quad (24)$$

PROOF. Define $\mathcal{E}_i = x_i^b - x_i^*$. The time derivative of $\mathcal{E}_i$ yields

$$\begin{aligned} \dot{\mathcal{E}}_i &= \dot{x}_i^b = \omega_{i,12}(x_i^a(t) - x_i^b(t)) \\ &= \omega_{i,12}(x_i^a - x_i^*) - \omega_{i,12}(x_i^b - x_i^*). \end{aligned} \quad (25)$$

Select Lyapunov function candidate

$$V = \frac{1}{2} \mathcal{E}_i^2. \quad (26)$$

The time derivative of V yields

$$\begin{aligned} \dot{V} &= \mathcal{E}_i \dot{\mathcal{E}}_i \\ &= \mathcal{E}_i [\omega_{i,12}(x_i^a - x_i^*) - \omega_{i,12}(x_i^b - x_i^*)] \\ &= -\omega_{i,12} \mathcal{E}_i^2 + \omega_{i,12} \mathcal{E}_i (x_i^a - x_i^*). \end{aligned} \quad (27)$$

Recall that $t \rightarrow \infty, x_i^a \rightarrow x_i^*, \forall i \in \mathcal{V}$. Thus, the system (25) is asymptotically stable. It obtains that $t \rightarrow \infty, x_i^b \rightarrow x_i^*, \forall i \in \mathcal{V}$.

Ultimately, based on the properties of integration functions, we conclude

$$\lim_{t \rightarrow \infty} x_i(t) = \lim_{t \rightarrow \infty} x_i^a(t) = \lim_{t \rightarrow \infty} x_i^b(t) = x_i^*, \forall i \in \mathcal{V}. \quad (28)$$

$\square$

3.4 Privacy-Preserving Property

Based on the state decomposition mechanism introduced in Section 2.4, the following privacy guarantee is established.

DEFINITION 2 (PRIVACY PRESERVATION [15, 18]). *In distributed game scenarios, the initial state $x_i(0)$ reflects the private preference or local information of agent i , whereas the Nash equilibrium is a collective outcome shared by all agents. Since all states asymptotically converge to the same equilibrium value, privacy protection in this work focuses on the initial state. Therefore, the system achieves privacy preservation if, for any regular agent i , no estimator constructed from the information available to the hostile agents can determine $x_i(0)$ within a bounded interval with certainty.*

THEOREM 4. *Under the state decomposition mechanism with unmodeled dynamics, the initial state $x_i^a(0)$ can be preserved under the control protocol (5) and (14).*

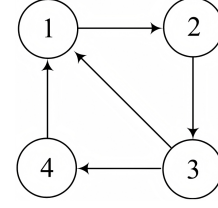


Figure 1: Communication topology diagram.

PROOF. The proof idea is analogous to [16], that is, to demonstrate that for any $\bar{x}_i(0)$ unequal to $x(0)$, the hostile agents observe the identical information.

Given varying initial values $x_i(0)$ and initial weights $w_{i,12}(0)$, $w_{i,21}(0)$, there are weights connecting the two substates $\bar{x}_i^a$ and $\bar{x}_i^b$ that satisfy the subsequent conditions:

- (1) If $|x_i^a(0) - x_i^b(0)| \leq \gamma_0$ and $|\bar{x}_i^a(0) - \bar{x}_i^b(0)| \leq \gamma_0$, then

$$\bar{w}_{i,21}(0) = \frac{w_{i,21}(0)(x_i^a(0) - x_i^b(0))}{\bar{x}_i^a(0) - \bar{x}_i^b(0)}.$$

- (2) If $|x_i^a(0) - x_i^b(0)| \leq \gamma_0$ and $|\bar{x}_i^a(0) - \bar{x}_i^b(0)| > \gamma_0$, then

$$\bar{w}_{i,21}(0) = \frac{w_{i,21}(0)|x_i^a(0) - x_i^b(0)|(x_i^a(0) - x_i^b(0))}{\Omega(0)(\bar{x}_i^a(0) - \bar{x}_i^b(0))}.$$

- (3) If $|x_i^a(0) - x_i^b(0)| > \gamma_0$ and $|\bar{x}_i^a(0) - \bar{x}_i^b(0)| \leq \gamma_0$, then

$$\bar{w}_{i,21}(0) = \frac{\gamma_0 w_{i,21}(0)(x_i^a(0) - x_i^b(0))}{|x_i^a(0) - x_i^b(0)|(\bar{x}_i^a(0) - \bar{x}_i^b(0))}.$$

- (4) If $|x_i^a(0) - x_i^b(0)| > \gamma_0$ and $|\bar{x}_i^a(0) - \bar{x}_i^b(0)| > \gamma_0$, then

$$\bar{w}_{i,21}(0) = \frac{w_{i,21}(0)|x_i^a(0) - x_i^b(0)|(x_i^a(0) - x_i^b(0))}{|x_i^a(0) - x_i^b(0)|(\bar{x}_i^a(0) - \bar{x}_i^b(0))}.$$

- (5) The weight $\bar{w}_{i,12}(0)$ satisfies

$$\bar{w}_{i,12}(0) = \frac{x_i^b(0) - \bar{x}_i^b(0) + w_{i,12}(0)(x_i^a(0) - x_i^b(0))}{\bar{x}_i^a(0) - \bar{x}_i^b(0)}.$$

Then, with the above design of the weight values, for any alternative initial state $\bar{x}_i(0)$ there exist private decomposition parameters that generate identical observable information to the hostile agents. Hence, based on their observations, the hostile agents cannot distinguish $x_i(0)$ from $\bar{x}_i(0)$, and the initial state privacy of agent i is preserved. $\square$

4 Numerical Example

In this section, a numerical simulation is presented to illustrate the effectiveness of the proposed results. We loop through three directed graphs in [24] that satisfy (2, 2)-robust which are shown in Fig 1, so the network is time-varying. Then the node 5 is chosen as adversarial agent and other nodes $i = [1, 2, 3, 4]$ represent regular ones.

In regard to our simplified second-order PTESO design in Section 3.1, the disturbance set of nodes is selected as $[d_1, d_2, d_3, d_4] =$

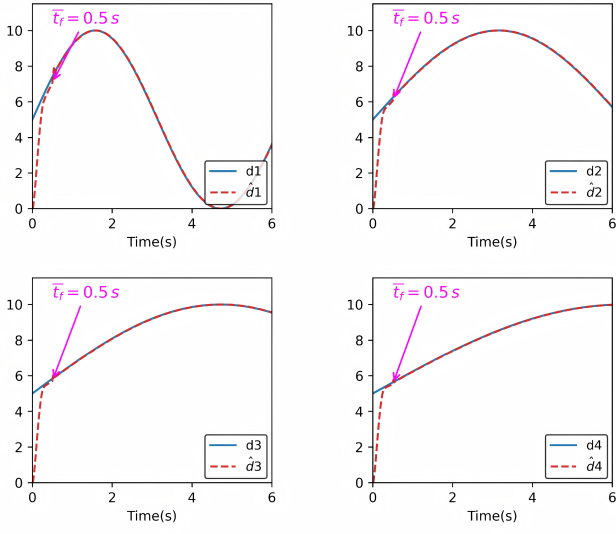


Figure 2: Trajectories of the disturbance d_i and the observations $\hat{d}_i$ of PTESO.

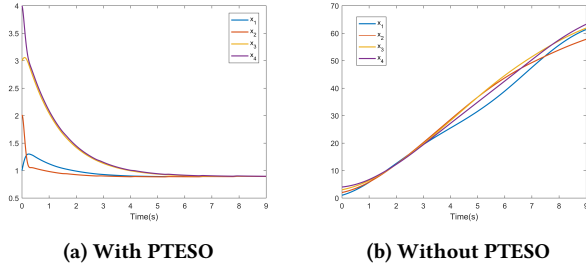


Figure 3: State trajectories x_i with and without PTESO compensation.

$[5 \sin(t) + 5, 5 \sin(\frac{t}{2}) + 5, 5 \sin(\frac{t}{3}) + 5, 5 \sin(\frac{t}{4}) + 5]$ under the condition of Remark 1. Moreover, set the prescribed convergence time $\bar{t}_f = 0.5s$, $\zeta = 4$, $l_1 = 6$, $l_2 = 10$, $\Gamma = 5$, $k_2 = 5$, $k_1 = 1$. As Fig. 2 shows, PTESO has high observation accuracy, and the peaking value problem has been alleviated.

The network weights $\omega_{ij}(t)$ and substates weights $\omega_{i,12}(t)$, $\omega_{i,21}(t)$ are both randomly selected in $[0.5, 1)$. The initial states which are considered as private information are set to $[x_1(0), \dots, x_4(0)] = [1, 2, 3, 4]$. At the initial moment $t = 0$, the initial state $x_i(0)$ is randomly decomposed into two substates $x_i^a(0)$, $x_i^b(0)$ that satisfy $x_i^a(0) + x_i^b(0) = 2x_i(0)$. And at all the next moments $t > 0$, $x_i(t)$ can be integrated by $x_i(t) = (x_i^a(t) + x_i^b(t))/2$ where $x_i^a(t)$, $x_i^b(t)$ are updated by aforementioned update laws. Set the step-size sequence $\gamma_k = 0.7^k$, where t satisfy $t * \Delta T = t$, and ΔT is the time interval of the discrete system.

As shown in Fig. 2, the proposed PTESO successfully followed the trajectories of the disturbance at $t = 0.5s$. And it can keep following after $0.5s$. Therefore, the proposed PTESO is proved to be valid. Next, we consider the condition of consensus. Under the

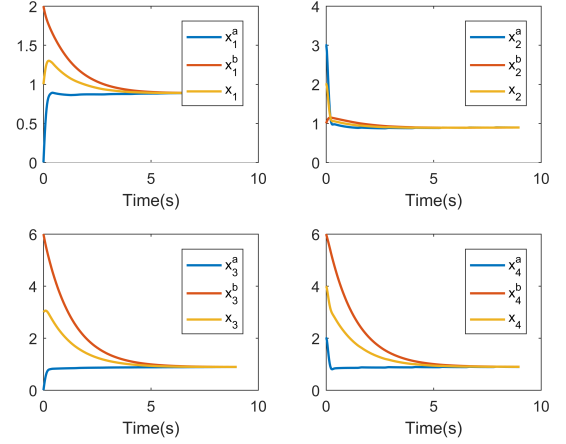
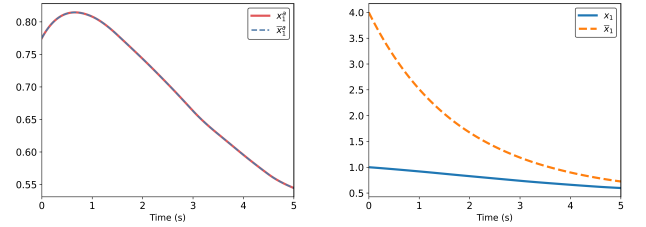


Figure 4: Evolution of substates and actual states x_i, x_i^a, x_i^b of regular nodes.



(a) Adversary-observable substate trajectories $x_1^a(t)$ and $\bar{x}_1^a(t)$. (b) Actual state trajectories $x_1(t)$ and $\bar{x}_1(t)$.

Figure 5: Adversary-view privacy validation.

influence of the disturbance d_i , as shown in Fig. 3(a), consensus cannot be achieved while the resilient consensus are reached, thanks to the anti-disturbance effect of PTESO in Fig. 3(b). In addition, two substates and actual state of regular nodes can also realize consensus as described in Fig. 4.

To supplement privacy verification, an adversary-perspective privacy verification is conducted under the same simulation settings. The adversary observes the publicly transmitted sub-state $x_i^a(t)$. The private sub-state $x_i^b(t)$ and internal coupling weight $\omega_{i,21}(t)$ remain inaccessible. Construct a set of distinct initial states $\bar{x}_i(0) \neq x_i(0)$, and adjust the corresponding private weights according to the construction in Theorem 4 such that the adversary observes $x_i^a(t)$ to be consistent with $\bar{x}_i^a(t)$. Fig. 5a presents the verification results, while Fig. 5b shows that the actual $x_1(t)$ and $\bar{x}_1(t)$ have different initial values and follow distinct paths. To highlight the privacy mechanism, this verification is conducted under the condition that PTESO had already effectively compensated for disturbances.

5 Conclusion

This paper studies the problem of distributed Nash equilibrium seeking in the presence of unmodeled dynamics in non-cooperative game networks. Based on a leaderless consensus protocol and gradient games, a Nash equilibrium seeking algorithm is designed and uses an expanded state observer with finite time convergence to estimate and compensate for unmodeled dynamics in the system. It is proved that in the presence of unmodeled dynamics in the system, the player's behavior can still asymptotically converge within the neighborhood of the Nash equilibrium. Simultaneously applying privacy protection to the initial state values to complete the indistinguishability construction. The theoretical results are verified through numerical simulations.

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